

CSE 306

Microprocessors, Micro-Controllers and Assembly Language Sessional

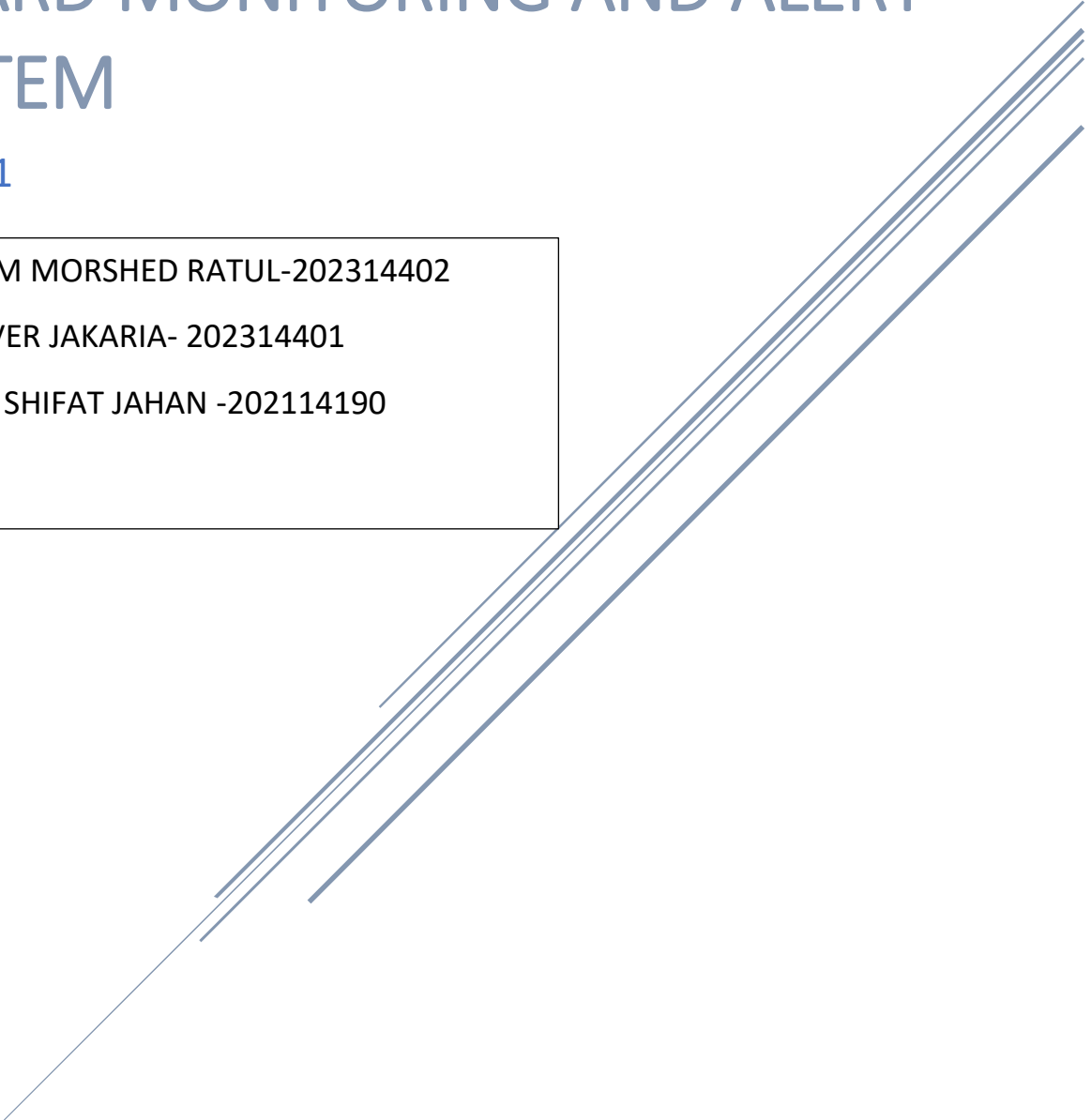
# GUARD MONITORING AND ALERT SYSTEM

Group 1

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# Project Name: Guard Monitoring and Alert System

## Project Description.

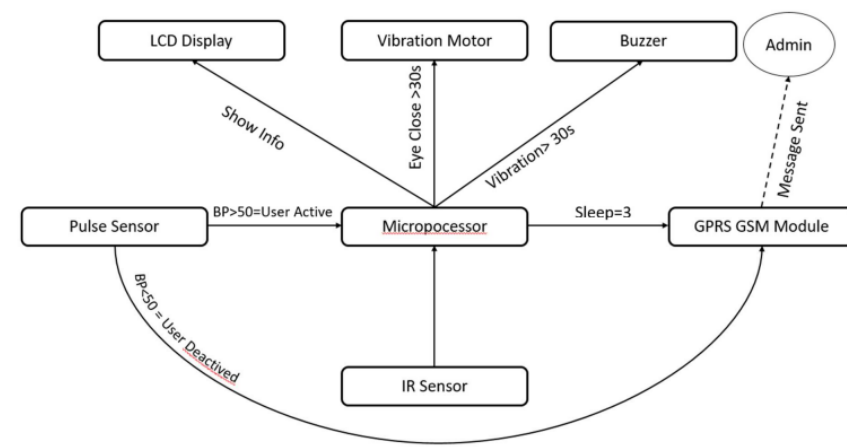
This project aims to enhance the efficiency and reliability of security personnel by continuously monitoring their alertness and physical well-being during duty. The system is designed to detect when a guard falls asleep, monitor their heart rate, and send alert messages to both the guard and the guard room via GSM.

## Objective:

1. Detect when a security guard falls asleep during duty using real-time monitoring.
2. Continuously monitor the guard's heart rate to ensure physical well-being.
3. Alert the guard immediately if drowsiness or abnormal vitals are detected.
4. Send real-time alert messages to the guard room via GSM communication.
5. Enhance the reliability and efficiency of security personnel on duty.

## Working Principle:

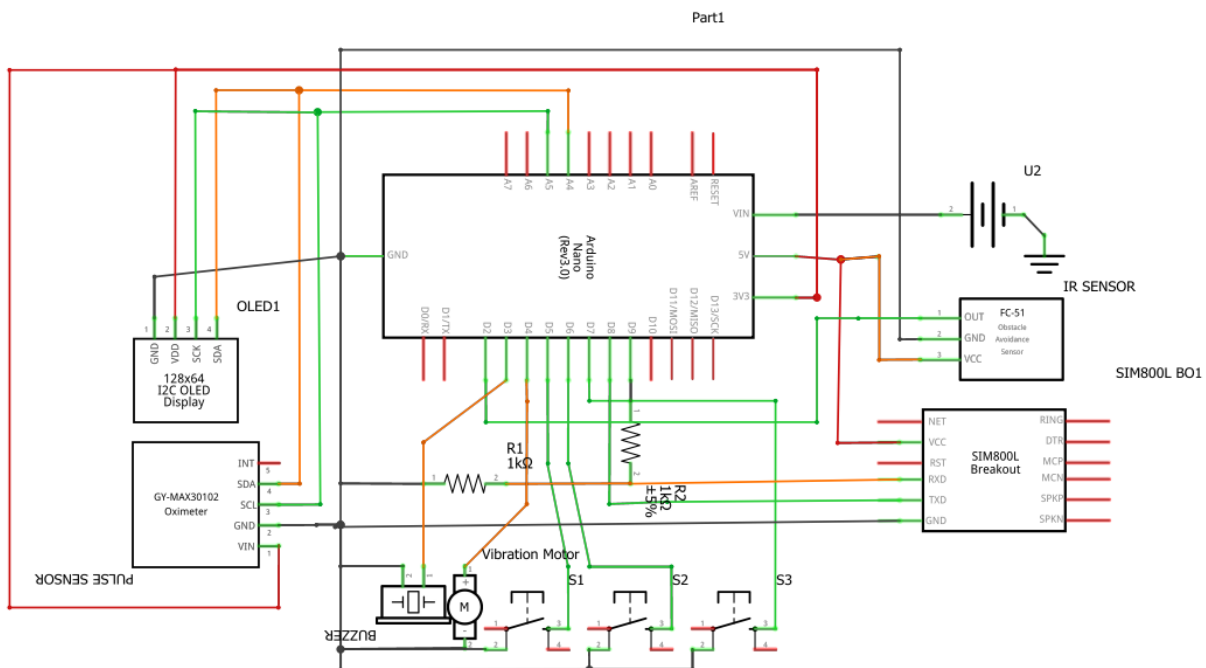
1. Eye Closure Detection: Using an IR sensor placed near the eye the system detects if the guard's eyes are closed. If the eyes remain closed for 30 seconds, it indicates the guard is likely asleep. Then the vibration motor will be on.
2. Local Wake-up System: Upon detecting sleep, the system activates a vibration module and buzzer will buzz. If buzzer is not off with in 1min then it will sent message to guard room as emergency.
3. Escalation Logic: If the guard is detected sleeping three consecutive times, the system sends a warning SMS to the guard room/control center to replace the guard.
4. SMS Alerts via GSM: Using a GSM/GPRS module (SIM900A), the system sends:
  - a. Initial Status updates when the guard is awake and pulse is normal.
  - b. Alerts when the guard sleeps more than three time.
  - c. Urgent messages for emergency.



## Component List with Model number and cost.

| Components                       | Quantity | Cost |
|----------------------------------|----------|------|
| Arduino Nano                     | 1        | 415  |
| 0.96" OLED Display module        | 1        | 395  |
| Jumper Wire                      | 20       | 110  |
| Miniature Vibration Motor 6x15   | 1        | 110  |
| Push Button                      | 3        | 21   |
| Sim800L Mini GPRS GSM Module     | 1        | 410  |
| IR Sensor TCRT 5000 dual Channel | 1        | 170  |
| Max30102 Pulse Oximeter          | 1        | 390  |
| Active Buzzer                    | 1        | 59   |
| 10k Potentiometer                | 3        | 39   |
| Silicon Wire                     | 1meter   | 60   |
| SPDT switch                      | 1        | 15   |
| JST pin connector                | 3        | 45   |
| Battery Charger                  | 1        | 40   |
| Lithium ion Batter 2200mAh 3.7V  | 1        | 710  |
| Bread Board                      | 1        | 110  |
| Total                            |          | 3099 |

## Schematic Diagram of the full circuit.



## **Future Implications/Improvements.**

5. The proposed system has significant potential for future development and improvement. with artificial intelligence (AI) algorithms for drowsiness and facial recognition. Such an upgrade would allow for more accurate and reliable monitoring, reducing false detections caused by sensor limitations.
6. Another area of improvement is the incorporation of wearable health monitoring devices, such as smart bands or helmets, which can measure not only heart rate but also blood pressure, oxygen levels, and fatigue indicators. This would enable a more comprehensive assessment of the guard's physical condition, thereby ensuring both safety and efficiency.
7. The implementation of Internet of Things (IoT) and cloud connectivity can also significantly expand the system's capabilities. By connecting to a centralized cloud server, data from multiple guards can be monitored in real time from the control room, while historical records can be analyzed to identify performance trends and health risks. Alongside this, a mobile application could be developed for supervisors, providing instant notifications, live status updates, and faster emergency response.
8. Finally, integration with biometric attendance systems such as fingerprint or RFID could ensure that guards are not only present but also actively monitored for alertness and well-being throughout their shift.